



Q4

Create a risk matrix (asset, threat, vulnerability, likelihood, impact, risk)

5 Marks

Objective: Build a simple risk matrix that rates each asset-threat pair by likelihood and impact and derives an overall risk level.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	List the key assets involved in the breach.	Asset inventory	Asset list
2	ANALYSE	For each asset, name the threat and the vulnerability it exploited.	Threat/vuln notes	Threat + vuln per asset
3	TEST	Rate likelihood as Low / Medium / High.	Likelihood scale	Likelihood set
4	TEST	Rate impact as Low / Medium / High.	Impact scale	Impact set
5	ANALYSE	Combine likelihood × impact to get the risk level and colour it.	Risk = L × I	Colour-coded risk
6	REPORT	Sort the matrix so the highest risks appear first.	Sort by risk	Prioritised matrix

Phase key: ● SETUP ● CONFIGURE ● EXECUTE ● TEST ● ANALYSE ● REPORT ● FINISH

Worked Example (with sample values & output)

Risk matrix (Acme HealthCare)

Asset	Threat	Vulnerability	Likelihood	Impact	Risk
Patient DB	Data theft	Weak email filter	High	High	CRITICAL
Booking portal	Ransomware	Unpatched server	Medium	High	HIGH
Staff accounts	Credential theft	No MFA	High	Medium	HIGH

Reading the matrix: the Patient DB is the top priority (High likelihood and High impact = Critical), so its mitigations are addressed first in Q5.



The Maersk "NotPetya" Cyberattack (2017).

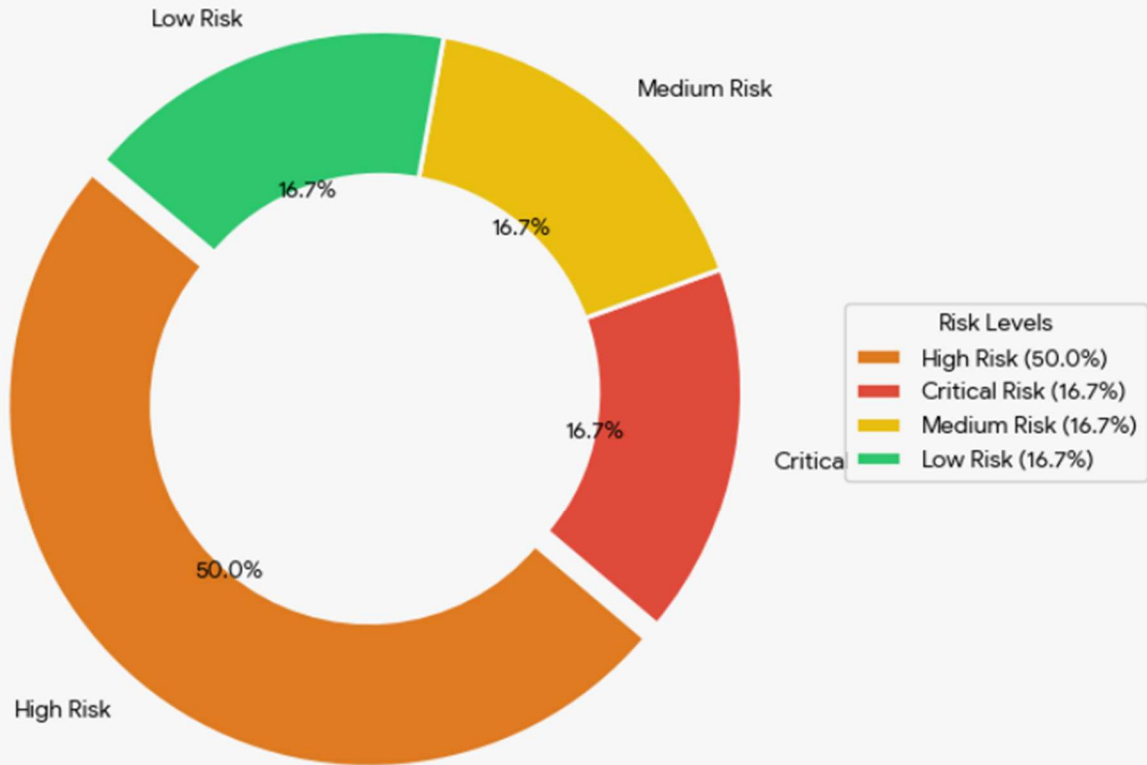
Asset	Threat	Vulnerability	Likelihood	Impact	Risk
Global Shipping Logistics & Booking Software	State-Sponsored Destructive Malware (NotPetya worm)	Supply chain vulnerability via unvalidated automatic updates of a trusted 3rd-party accounting tool.	Medium	Critical	 Critical Risk
Maersk's centralized IT network, active directory, and global container booking systems managing roughly 15-18% of all global ocean freight	A highly aggressive, self-propagating cyber weapon disguised as ransomware but built solely to wipe data and destroy networks.	Maersk used a local tax software package called <i>M.E.Doc</i> in their Ukraine office. Russian military hackers hijacked <i>M.E.Doc's</i> update servers, injecting malware into a routine software update. Because Maersk's network trusted the vendor's software updates implicitly and lacked internal network segmentation, the malware spread globally within hours.	Medium	Critical	 Critical Risk

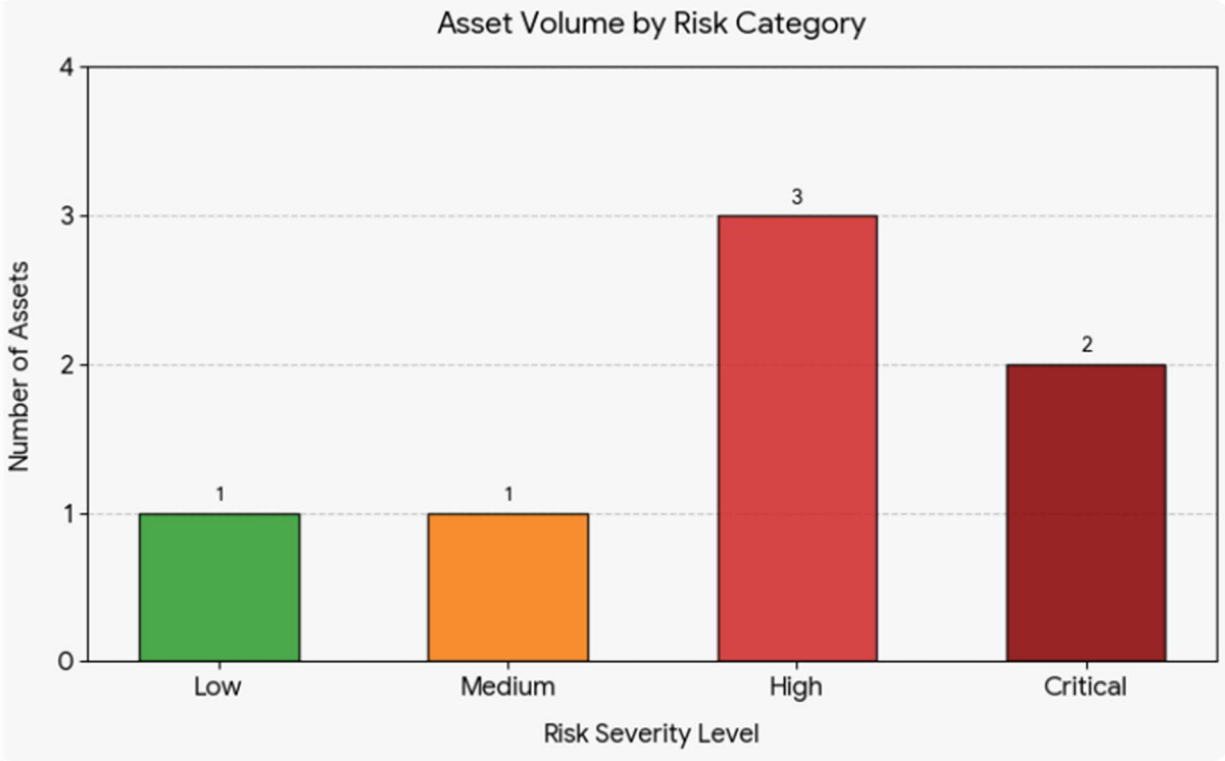
Your Output / Screenshot (paste result here):

Maersk was forced to halt operations across **76 global port terminals**. Ships were stranded at sea, cargo could not be unloaded, and the company had to rebuild its entire IT infrastructure from scratch over two grueling weeks. The disaster cost Maersk between **\$250 million and \$300 million** in direct revenue losses.



Risk Level Distribution across Assets





- **Critical Risk (1 Asset):** Customer Database (High Likelihood × Critical Impact)
- **High Risk (3 Assets):** E-commerce Website, Internal Source Code, Company Email System
- **Medium Risk (1 Asset):** Office Laptops (Medium Likelihood × Medium Impact)
- **Low Risk (1 Asset):** Power Supply Server Room (Low Likelihood × Medium Impact).